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Technical Skills

Languages TypeScript, Python	Frontend React, Next.js, Tailwind CSS	Backend Node.js, Express.js, REST APIs, Event-Driven Architecture, Async Processing, JWT, RBAC
Databases PostgreSQL, Redis, MongoDB	Async & Messaging BullMQ, Dead-Letter Queues, Idempotent Processing	Infrastructure Docker, GitHub Actions CI/CD, AWS ECS Fargate, Vercel, DigitalOcean

Work Experience

Vogel Edits 01/2025 - Present
Full-Stack Engineer (Adobe Extensions)

- Shipped production **React** Adobe After Effects & Premiere Pro extensions, including Vanish BG (real-time AI background removal) running ML inference inside a sandboxed CEP environment and distributed via aescrpts.com.
- Designed a **Node.js** sidecar architecture to bypass CEP runtime limitations, using local HTTP IPC and streaming inference results back to the UI in real time.
- Built a Premiere Pro TTS extension with local **ONNX** inference, delivering an end-to-end pipeline from text input to rendered timeline audio clips.
- Collaborated asynchronously with designers and stakeholders using agile workflows to deliver production-ready tools.

Freelance 11/2024 - 01/2025
Full-Stack Engineer

- Built a full-stack tourism booking platform with **Next.js** and **TypeScript**, implementing a **Node.js** API with type-safe **PostgreSQL** access via **TypeORM**.
- Drove API performance using **PostgreSQL** indexing and **EXPLAIN ANALYZE**, achieving sub-100ms response times under concurrent load.
- Integrated Paymob payments with **idempotent** transaction handling and webhook verification, reducing checkout friction by 40%.
- Improved SEO and Core Web Vitals to 95+ Lighthouse via **Next.js SSR**, image optimization, and lazy loading.

Projects

Medical CRM — AI-Enhanced Prescription Platform
Tech Stack: React, Node.js, PostgreSQL

- Built a full-stack CRM for diagnoses, prescriptions, and patient records using **React**, **Node.js**, and **PostgreSQL**.
- Implemented secure auth with **JWT** access/refresh token rotation and **RBAC** across three permission levels.
- Designed drug interaction validation across 500+ medications using graph-based **PostgreSQL** lookups.
- Optimized dashboard queries for 10K+ monthly records using composite indexes and window functions.

RootCluster — AI-Powered Log Analysis Platform <https://rootcluster.dev>
Tech Stack: Node.js, TypeScript, PostgreSQL (pgvector), Redis, BullMQ, ONNX, AWS ECS Fargate, Docker, GitHub Actions

- Designed and deployed a distributed, **event-driven** log ingestion pipeline on **AWS ECS Fargate** using **BullMQ** and **Redis**, with **PostgreSQL (pgvector)** similarity search, processing 1M+ events/hour with sub-second latency.
- Implemented robust **async processing** with producer-consumer queues, configurable worker concurrency, and priority queues for urgent events vs. backfills.
- Built **dead-letter queues** with retry policies (exponential backoff, max attempts) to isolate malformed entries without blocking the pipeline.
- Designed **idempotent** processing using Redis dedup keys and at-least-once delivery semantics via job completion callbacks.
- Reduced **ONNX** embedding time from 20+ minutes to under 1 minute via batch processing, Redis caching, and

worker autoscaling.

- Set up CI/CD using **GitHub Actions** with OIDC, Docker image publishing, and AWS integration for secrets and deployments.

Coursework

- Meta Certified Frontend Developer — Meta (2024)
- Complete SQL Databases Bootcamp: Zero to Mastery — Udemy
- Master the Coding Interview: Data Structures & Algorithms — Udemy
- Node.js: The Complete Guide — Udemy
- Grokking Algorithms — Aditya Bhargava
- Designing Data-Intensive Applications — Martin Kleppmann